# Linear Algebra

Revision Sheet

Every topic on its own page • formulas, method, worked example, traps

## How to use this sheet

- **One topic per page.** Nothing splits across pages — read a page, close the book, and try to reproduce the formula box from memory.
- **Read in this order on each page:** the italic line (what the topic *is*) → the formula box → "how to approach" → the worked example → the traps.
- **The dashed "My notes" box is yours.** Add the variant that caught you out, a formula your coaching notes state differently, or the page number of a question you got wrong.
- **Traps in the orange box are the marks people actually lose.** Re-read only those the night before the exam.

## Contents


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# 1. Matrix Basics & Operations

*Nothing here is hard, but the transpose/inverse reversal rule and non-commutativity generate a surprising number of wrong answers.*

## KEY FORMULAS

**Product**

$$(AB)_{ij} = \sum_k A_{ik}B_{kj} \quad (m \times n)(n \times p) = (m \times p)$$

**Order reverses**

$$(AB)^\mathsf{T} = B^\mathsf{T}A^\mathsf{T}, \qquad (AB)^{-1} = B^{-1}A^{-1}$$

**Inverse exists iff**

$$\det A \neq 0$$

**Dot product and length**

$$x \cdot y = x^\mathsf{T}y, \qquad \|x\|_2 = \sqrt{x^\mathsf{T}x}$$

## WHAT YOU MUST KNOW

- $AB \neq BA$ in general, but $(AB)C = A(BC)$ and $A(B+C) = AB + AC$ always hold.
- $AB = 0$ does *not* imply $A = 0$ or $B = 0$.
- $AB = AC$ does *not* imply $B = C$ unless $A$ is invertible.
- For $2 \times 2$: $A^{-1} = \dfrac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

## HOW TO APPROACH A QUESTION

1. Check dimensions first — most "impossible" options fail on shape alone.
2. For $(\cdot)^\mathsf{T}$ or $(\cdot)^{-1}$ of a product, **reverse the order**.
3. Before inverting, confirm $\det \neq 0$.

## WORKED EXAMPLE

**Q.** If $A$ is $3 \times 4$ and $B$ is $4 \times 2$, what is the shape of $(AB)^\mathsf{T}$?
$AB$ is $3 \times 2$, so $(AB)^\mathsf{T}$ is $\mathbf{2 \times 3}$ — equivalently $B^\mathsf{T}A^\mathsf{T} = (2 \times 4)(4 \times 3)$.

## WATCH OUT

- Writing $(AB)^\mathsf{T} = A^\mathsf{T}B^\mathsf{T}$ — the order must reverse.
- Cancelling matrices as if they were numbers.

## MY NOTES

# 2. Trace, Determinant & Their Identities

*Trace and determinant are the two numbers that connect a matrix to its eigenvalues without ever solving for them.*

## KEY FORMULAS

**Trace**

$$\operatorname{tr}(A) = \sum_i A_{ii} = \sum_i \lambda_i$$

**Cyclic property**

$$\operatorname{tr}(AB) = \operatorname{tr}(BA)$$

**Determinant and eigenvalues**

$$\det(A) = \prod_i \lambda_i$$

**Multiplicative**

$$\det(AB) = \det A \det B, \qquad \det(A^{-1}) = \frac{1}{\det A}$$

**Scaling an $n \times n$ matrix**

$$\det(cA) = c^n \det A$$

## WHAT YOU MUST KNOW

- $\det(A^{\mathsf{T}}) = \det A$; for triangular or diagonal matrices det = product of the diagonal.
- Row swap flips the sign; adding a multiple of one row to another leaves det unchanged.
- $\det A = 0 \iff$ singular $\iff$ rows/columns dependent $\iff$ 0 is an eigenvalue.
- **Use trace and determinant to find eigenvalues of a $2 \times 2$ without the characteristic polynomial:** $\lambda_1 + \lambda_2 = \text{tr}$, $\lambda_1 \lambda_2 = \det$.

## HOW TO APPROACH A QUESTION

1. If the question gives you tr and det, you already have the eigenvalue sum and product.
2. Never expand a large determinant — row-reduce to triangular form first.
3. Watch for $\det(cA) = c^n \det A$; the exponent $n$ is the *size*, not the scalar.

## WORKED EXAMPLE

**Q.** $A$ is $3 \times 3$ with $\det A = 4$. Find $\det(2A)$ and $\det(A^{-1})$.

$$\det(2A) = 2^3 \cdot 4 = \mathbf{32}, \qquad \det(A^{-1}) = \tfrac{1}{4} = \mathbf{0.25}$$

## WATCH OUT

- Using $\det(cA) = c \det A$ — it is $c^n$.
- Assuming $\det(A + B) = \det A + \det B$ (false).

## MY NOTES

# 3. Rank, Nullity & Linear Systems

*One picture answers every 'how many solutions?' question: compare the rank of $A$ with the rank of the augmented matrix, then with $n$.*

## KEY FORMULAS

**Rank–nullity (columns!)**

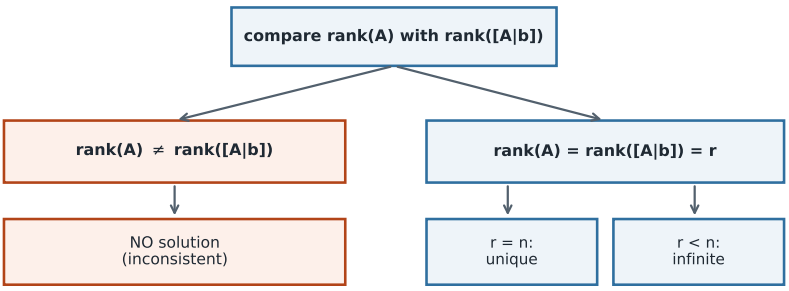
$$\operatorname{rank}(A) + \operatorname{nullity}(A) = n$$

**Consistency of $Ax = b$**

$$\text{consistent} \iff \operatorname{rank}(A) = \operatorname{rank}([A \,|\, b])$$

## WHAT YOU MUST KNOW

- Rank = number of non-zero rows after row reduction = independent rows = independent columns.
- $Ax = 0$ has a **non-trivial** solution iff $\operatorname{rank}(A) < n$.
- $\operatorname{rank}(A) \le \min(m, n)$; $\operatorname{rank}(AB) \le \min(\operatorname{rank}A, \operatorname{rank}B)$.
- For a square $A$: invertible $\iff$ rank $= n$ $\iff$ $\det \neq 0$ $\iff$ nullity $= 0$.




## HOW TO APPROACH A QUESTION

1. Row-reduce $[A \,|\, b]$ once — it gives both ranks at the same time.
2. Compare the two ranks: unequal $\Rightarrow$ no solution.
3. If equal, compare with $n$ (the number of *unknowns*): $= n$ unique, $< n$ infinitely many.
4. Number of free variables $= n -$ rank.

## WORKED EXAMPLE

**Q.** $A$ is $3 \times 5$ with $\operatorname{rank}(A) = 3$. How many free variables does $Ax = 0$ have?

$$\text{nullity} = n - \text{rank} = 5 - 3 = \mathbf{2}$$

## WATCH OUT

- Using the number of *rows* in rank–nullity — it is always the number of columns.
- Forgetting that a consistent system can still have infinitely many solutions.

## MY NOTES

# 4. Vector Spaces, Independence & Basis

*Independence, spanning and dimension are three views of the same question: how much information do these vectors actually carry?*

## KEY FORMULAS

**Independence**

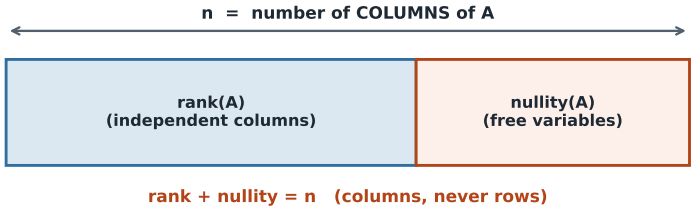
$$\sum_i c_i v_i = 0 \;\Rightarrow\; \text{all } c_i = 0$$

**Dimension**

$$\dim(\text{col space}) = \text{rank}, \qquad \dim(\text{null space}) = \text{nullity}$$

## WHAT YOU MUST KNOW

- A basis is an independent *and* spanning set; every basis of a space has the same size.
- $n$ vectors in $\mathbb{R}^n$ form a basis $\iff$ the matrix they form is invertible.
- More than $n$ vectors in $\mathbb{R}^n$ are **always** dependent.
- Any set containing the zero vector is dependent.




## HOW TO APPROACH A QUESTION

1. Stack the vectors as columns and row-reduce; rank = number of independent ones.
2. If #vectors > dimension $\Rightarrow$ dependent immediately, no computation needed.
3. For "do they span $\mathbb{R}^n$?" check rank $= n$.

## WORKED EXAMPLE

**Q.** Are $(1,2),(2,4),(1,0)$ independent in $\mathbb{R}^2$?
Three vectors in a 2-dimensional space are **always dependent** — no calculation required. (Indeed $(2,4) = 2(1,2)$.)

## WATCH OUT

- Testing independence by checking pairs only — dependence can involve three or more vectors.
- Confusing the dimension of the space with the number of vectors.

## MY NOTES

# 5. Eigenvalues & Eigenvectors

*GATE DA rarely makes you grind out eigenvalues. It gives you a condition like $A^2=A$ and asks what that forces.*

## KEY FORMULAS

**Definition**

$$Av = \lambda v, \qquad v \neq 0$$

**Characteristic equation**

$$\det(A - \lambda I) = 0$$

**Two free checks**

$$\sum_i \lambda_i = \operatorname{tr}(A), \qquad \prod_i \lambda_i = \det(A)$$

**Transformed matrices**

$$A^k \to \lambda^k, \quad A^{-1} \to \tfrac{1}{\lambda}, \quad A + cI \to \lambda + c$$

## WHAT YOU MUST KNOW

- $A^2 = A$ **(idempotent)** $\Rightarrow \lambda \in \{0, 1\}$, and $\operatorname{tr}(A) = \operatorname{rank}(A)$.
- $A^3 = A \Rightarrow \lambda \in \{-1, 0, 1\}$. $A^k = 0$ **(nilpotent)** $\Rightarrow$ all $\lambda = 0$.
- **Symmetric** $A$: eigenvalues real, eigenvectors orthogonal. **Orthogonal** $Q$: $|\lambda| = 1$.
- $M = xx^{\mathsf{T}}$: rank 1, symmetric, one non-zero eigenvalue $= x^{\mathsf{T}}x$, rest zero.

## HOW TO APPROACH A QUESTION

1. Read the *condition* first — "$A^2 = A$", "$\|Ax\| = \|x\|$" — and write down what it forces about $\lambda$.
2. Only if no condition is given, form $\det(A - \lambda I) = 0$.
3. Cross-check your eigenvalues against tr and det before answering.

## WORKED EXAMPLE

**Q.** $A$ is $4 \times 4$, $A^2 = A$, $\operatorname{rank}(A) = 3$. Find $\operatorname{tr}(A)$ and the eigenvalues.
Idempotent $\Rightarrow$ each $\lambda$ is 0 or 1, and the number of 1s equals the rank.

$$\lambda = 1, 1, 1, 0 \quad \Rightarrow \quad \operatorname{tr}(A) = \mathbf{3}$$

## WATCH OUT

- Solving the characteristic polynomial when a one-line structural fact was intended.
- Forgetting that eigenvalues of $A^{-1}$ are $1/\lambda$, not $-\lambda$.

## MY NOTES

# 6. Special Matrix Types as Properties

*MSQ questions here never care about the entries — only about the defining identity. Learn the identity and its two consequences.*

## KEY FORMULAS

**Orthogonal**

$$Q^{\mathsf{T}}Q = I, \quad Q^{-1} = Q^{\mathsf{T}}, \quad \|Qx\| = \|x\|, \quad \det Q = \pm 1$$

**Idempotent / projection**

$$P^2 = P; \quad \text{projection also } P^{\mathsf{T}} = P$$

**Projection trace**

$$\operatorname{tr}(P) = \operatorname{rank}(P) = \dim(\text{subspace})$$

**Gram matrix**

$$G = X^{\mathsf{T}}X \text{ is symmetric PSD}$$

## WHAT YOU MUST KNOW

- $X^{\mathsf{T}}X$ is invertible (and positive definite) $\iff$ the columns of $X$ are independent.
- $A = I + xx^{\mathsf{T}}$ with $\|x\| = 1$: eigenvalues are 2 (once) and 1 ($n-1$ times).
- Symmetric $\Rightarrow$ orthogonally diagonalisable: $A = Q\Lambda Q^{\mathsf{T}}$.
- Orthogonal matrices preserve lengths *and* angles — rotations and reflections.

## HOW TO APPROACH A QUESTION

1. Identify the defining identity in the question stem.
2. Write down the eigenvalue consequence and the trace/rank consequence.
3. Test each MSQ option against those two facts — never against sample entries.

## WORKED EXAMPLE

**Q.** $P$ is a projection onto a 2-dimensional subspace of $\mathbb{R}^5$. Give $\operatorname{tr}(P)$ and its eigenvalues.

$$\operatorname{tr}(P) = \operatorname{rank}(P) = \mathbf{2}; \qquad \lambda = 1, 1, 0, 0, 0$$

## WATCH OUT

- Assuming every symmetric matrix is a projection (it needs $P^2 = P$ too).
- Thinking $\det Q = 1$ for all orthogonal $Q$ — reflections give $-1$.

## MY NOTES

# 7. Diagonalization

*Diagonalization turns matrix powers into number powers. The only real question is whether enough independent eigenvectors exist.*

## KEY FORMULAS

**Decomposition**

$$A = PDP^{-1}, \qquad D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$$

**Powers become cheap**

$$A^k = PD^kP^{-1}$$

**Symmetric case (spectral theorem)**

$$A = Q\Lambda Q^{\mathsf{T}}, \quad Q \text{ orthogonal}$$

## WHAT YOU MUST KNOW

- $A$ is diagonalizable $\iff$ it has $n$ linearly independent eigenvectors.
- $n$ *distinct* eigenvalues $\Rightarrow$ diagonalizable (sufficient, not necessary).
- Every real symmetric matrix is diagonalizable — always.
- Columns of $P$ are the eigenvectors, in the same order as the $\lambda$s in $D$.

## HOW TO APPROACH A QUESTION

1. Find the eigenvalues; if all distinct, stop — it is diagonalizable.
2. If an eigenvalue repeats, check whether its eigenspace has full multiplicity.
3. For $A^k$ questions, diagonalize rather than multiplying repeatedly.

## WORKED EXAMPLE

**Q.** $A = PDP^{-1}$ with $D = \operatorname{diag}(1, 2)$. Find the eigenvalues of $A^{10}$.

$$D^{10} = \operatorname{diag}(1^{10}, 2^{10}) = \operatorname{diag}(1, 1024) \;\Rightarrow\; \lambda = \mathbf{1},\ \mathbf{1024}$$

## WATCH OUT

- Assuming a repeated eigenvalue means not diagonalizable — check the eigenspace.
- Mismatching the order of eigenvectors in $P$ with the eigenvalues in $D$.

## MY NOTES

# 8. LU Decomposition

*LU is just Gaussian elimination, bookkept: U is what you end up with, L records the multipliers you used.*

## KEY FORMULAS

**Factorisation**

$$A = LU \qquad \text{(with pivoting } PA = LU\text{)}$$

**Solve in two easy steps**

$$Ax = b \;\Rightarrow\; Ly = b \text{ then } Ux = y$$

**Determinant**

$$\det A = \prod_i U_{ii} \quad \text{(unit-diagonal } L\text{)}$$

## WHAT YOU MUST KNOW

- $L$ is lower-triangular with 1s on the diagonal (Doolittle); $U$ is upper-triangular.
- $L_{ij}$ is exactly the multiplier used to eliminate the $(i, j)$ entry.
- Forward substitution ($Ly = b$) then back substitution ($Ux = y$) — each is $O(n^2)$.
- An $LU$ without pivoting exists when every leading principal minor is non-zero.

## HOW TO APPROACH A QUESTION

1. Row-reduce $A$ to upper-triangular; that result *is* $U$.
2. Each time you do $R_i \to R_i - m\,R_j$, record $m$ in position $L_{ij}$.
3. Verify by multiplying $LU$ back — it must reproduce $A$ exactly.

## WORKED EXAMPLE

**Q.** Factor $A = \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix}$.

$R_2 \to R_2 - 2R_1$ gives $U = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$, and the multiplier 2 goes into $L_{21}$:

$$L = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}, \qquad U = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$$

## WATCH OUT

- Putting the multiplier in with the wrong sign — $L_{21}$ is $+2$, not $-2$.
- Row-swapping without recording it (that needs the $P$ in $PA = LU$).

## MY NOTES

# 9. SVD, Norms & Least Squares

*SVD works for any matrix, square or not. It is also the bridge from linear algebra to PCA in the ML paper.*

## KEY FORMULAS

**Decomposition**

$$A = U\Sigma V^{\mathsf{T}}$$

**Singular values**

$$\sigma_i = \sqrt{\lambda_i(A^{\mathsf{T}}A)} \;\geq 0$$

**Matrix norms**

$$\|A\|_2 = \sigma_{\max}, \qquad \|A\|_F^2 = \sum_i \sigma_i^2 = \sum_{ij} A_{ij}^2$$

**Least squares (normal equations)**

$$\hat{\beta} = (X^{\mathsf{T}}X)^{-1}X^{\mathsf{T}}y$$

## WHAT YOU MUST KNOW

- Singular values are **never negative**; the number of non-zero $\sigma_i$ equals the rank.
- Vector norms: $\|x\|_1 = \sum |x_i|$, $\|x\|_2 = \sqrt{\sum x_i^2}$, $\|x\|_\infty = \max |x_i|$.
- PCA link: eigenvalue of the covariance $= \sigma_i^2/n$ for centred data.
- For $M = xx^{\mathsf{T}}$ there is a single non-zero singular value $= \|x\|_2^2$.

## HOW TO APPROACH A QUESTION

1. For singular values, compute the eigenvalues of $A^{\mathsf{T}}A$ and take positive square roots.
2. For $\|A\|_F$ it is usually faster to square and add all entries directly.
3. For least squares, check $X^{\mathsf{T}}X$ is invertible (independent columns) first.

## WORKED EXAMPLE

**Q.** $u = (1, 2, 3)^{\mathsf{T}}$, $M = uu^{\mathsf{T}}$. Find $\sum_i \sigma_i$.

$M$ has rank 1, so only one non-zero singular value, equal to $u^{\mathsf{T}}u$:

$$\sum \sigma_i = 1^2 + 2^2 + 3^2 = \mathbf{14}$$

## WATCH OUT

- Reporting a negative singular value — impossible by definition.
- Confusing eigenvalues of $A$ with singular values of $A$ (equal only for symmetric PSD $A$).

## MY NOTES

# 10. Quadratic Forms & Positive Definiteness

*This is where linear algebra meets optimization and ML: definiteness is exactly convexity of the quadratic.*

## KEY FORMULAS

**Quadratic form**

$$q(x) = x^{\mathsf{T}}Ax \qquad (A \text{ symmetric})$$

**Positive definite**

$$x^{\mathsf{T}}Ax > 0 \;\; \forall x \neq 0 \iff \text{all } \lambda_i > 0$$

**Positive semidefinite**

$$x^{\mathsf{T}}Ax \geq 0 \iff \text{all } \lambda_i \geq 0$$

**Sylvester's criterion**

$$\text{PD} \iff \text{all leading principal minors} > 0$$

## WHAT YOU MUST KNOW

- $X^{\mathsf{T}}X$ is always PSD; it is PD exactly when the columns of $X$ are independent.
- $q$ is **convex** $\iff$ $A$ is PSD — the direct link to the optimization paper.
- A PD matrix is invertible and all its diagonal entries are positive.
- If $A$ is PD then so is $A^{-1}$.

## HOW TO APPROACH A QUESTION

1. Symmetrise first if the matrix given is not symmetric: use $\frac{1}{2}(A + A^{\mathsf{T}})$.
2. For $2 \times 2$, PD $\iff$ $a > 0$ *and* $\det > 0$ — fastest possible check.
3. For larger matrices, eigenvalue signs or leading minors, whichever is quicker.

## WORKED EXAMPLE

**Q.** Is $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ positive definite?

$$a_{11} = 2 > 0, \qquad \det = 4 - 1 = 3 > 0 \;\Rightarrow\; \textbf{yes, PD}$$

Check: eigenvalues are tr$\pm$ ... $= 3$ and $1$, both positive.

## WATCH OUT

- Testing definiteness on a non-symmetric matrix without symmetrising.
- Checking only $\det > 0$ — $\begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix}$ has $\det = 4$ but is negative definite.

## MY NOTES